Mai Khoi **Tieu**

Data Scientist | Analytics Engineer

linkedin.com/in/tieukhoimai github.com/tieukhoimai
tieukhoimai@gmail.com
Ho Chi Minh City, Vietnam
Female | Born 1998 | Vietnamese



I hold an M.Sc. in Data Science from the University of Verona, Through the Erasmus program, I spent a transformative semester at the University of Oslo. Since April 2025, I have been a research intern with the Trustworthy Green IoT Group at SINTEF Digital in Norway. Previously, I spent four years working as a Data Analyst/Engineer in Vietnam on large-scale IoT products, including ADAS radar systems at Bosch and WearOS smartwatch platforms at Fossil Group. My expertise spans analytics engineering and AI research, particularly in delivering intelligent monitoring and multi-agent systems for IoT and Digital Twins. I am proficient in LLM/agent frameworks, evaluation suites, CI/CD processes, and low-latency services.

Education

2023-2025	M.Sc. in Data Science, University of Verona *(CGPA 108/110)*
2024-2025	Erasmus Program, University of Oslo
2017-2021	B.Sc. in Mathematics, University of Science, VNUHCM (Grade: A - Thesis Grade: 9.5/10)
2016-2020	B.B.A. in Business Administration, Ho Chi Minh City Open University (CGPA: 3.40/4.0 - Ranked 4th overall)

Work Experience

Aug 2026 – Present — **PhD Candidate**, *NTNU*, *Trondheim, Norway*

- Conducting research on Agentic AI at Faculty of Information Technology and Electrical Engineering (IE) on multi-agent communication protocols for enhancing human learning.

Agentic AI | Human Computer Interaction (HCI) | Learning Technology

Apr 2025 – Sep 2025 — **Research Intern**, *SINTEF*, *Oslo, Norway*

- Advanced the SINDIT Digital Twin by extending its knowledge-graph schema and improving interoperability across sensor streams, asset metadata, and system ontologies.
- Developed a web-based interface for real-time visualization and interaction with the knowledge graph and digital twin state.
- Designed and built an LLM-enhanced multi-agent architecture enabling real-time monitoring, contextual retrieval, and explainable human-in-the-loop assistance on live factory data.

Knowledge Graphs | Neo4j | GraphDB | SvelteKit | d3.js | RAG | LangChain | LLMs | Multi-Agent Systems

Apr 2022 – Sep 2023 — **Analytics Project Manager & Data Analyst**, *Fossil Group Inc.*, *Ho Chi Minh City, Vietnam*

- Promoted and held a dual role as Analytics Project Manager and Data Analyst in the 2nd year.
- Planned sprint resources, timelines, and schedules for the analytics section in internal program release.
- Led cross-functional teams to develop an A/B testing workflow for wearable devices and companion apps (experiment design, wireframes, rollout).
- Implemented metrics frameworks and dynamic dashboards to track KPIs and gain valuable insights into user behavior, device performance monitoring, and wellness data from wearable devices.
- Designed event tracking and data flows between IoT devices and apps.
- Developed an internal Tracking Event Management tools for Data Quality Monitoring and Self-service Analytics, which automates the generation of daily queries and implements result alerts.

BigQuery | Python | Looker | Tableau | AWS | Redash | Grafana | Firebase | IoT | Wearable Devices | Project Management

Aug 2021 – Feb 2022 — **Data Analyst**, *Amanotes*, *Ho Chi Minh City, Vietnam*

- Performed statistical and analytical analyses, including correlation between app events, A/B testing, cohort analysis, and aha-moment identification in the music game.
- Built and maintained data pipelines with automated tests.
- Implemented and analyzed spectrograms of musical notes of another instrument.
- Trained and optimized TensorFlow Lite models for mobile inference.

BigQuery | Vertex AI | Airflow | dbt | Python | Metabase | A/B Testing

Mar 2020 – Aug 2021 — **Software Engineer, Data**, *Bosch Global Software Technologies*, *Ho Chi Minh City, Vietnam*

- Collaborated with Bosch Automotive Products on automated driving use cases and data quality.
- Analyzed software development logs and automotive radar data against QA/validation metrics for Advanced Driver Assistance Systems and Automotive Software Development Validation.
- Built and maintained analytics CI pipelines in Jenkins (automated jobs, scheduled reporting).
- Developed and operated data pipelines, databases, and dashboards for engineering stakeholders.
- Implemented automated database deployment and dashboard releases to validation workflows.

Jenkins | SQL | Python | Power BI | Dash | CI/CD | ADAS | Radar

Publications

Journal Articles

1. Tieu, M. K., H. V. Pham, N. T. Vu, and P. T. H. Nguyen (2022). "STL Decomposition-Based Stock Price Forecasting Using LSTM Model". In: *Journal of Data Science and Artificial Intelligence* 1.1.
2. Vuong, P. H., T. T. Dat, M. K. Tieu, P. H. Uyen, and P. T. Bao (2022). "Stock-Price Forecasting Based on XGBoost and LSTM". In: *Computer Systems Science and Engineering* 40.1, pp. 237–24.

Conference Proceedings

1. Lam, An Ngoc, Gøran Brekke Svaland, Volker Hoffmann, Mai Khoi Tieu, Brian Elvesæter, Xiang Ma, Dumitru Roman, and Phu H Nguyen (2025). "A Data Space-Driven Approach for Interoperability in Federated Digital Twins". In: *International Conference on Management of Digital*. Springer, pp. 192–206.
2. Tieu, Mai Khoi, Phu H Nguyen, An Ngoc Lam, and Sagar Sen (2025). "Multi-agent Generative AI on Digital Twins: A Learning Feedback Loop Framework for Monitoring and Explainability". In: *International Conference on Management of Digital*. Springer, pp. 72–86.

Skills

Programming Languages	Python, JavaScript/TypeScript, C/C++, MATLAB, R
Web & Frontend	React, Next.js, Tailwind CSS
AI/ML & LLMs	TensorFlow, PyTorch, LangChain, Hugging Face, RAG
Data & Pipelines	Airflow, dbt, SQL (PostgreSQL), BigQuery, MongoDB
Graph & Vector DBs	Neo4j (Cypher), GraphDB, Pinecone
Cloud & DevOps	GCP (BigQuery), AWS (S3, Redshift), Docker, Git/GitHub
Tools	Jenkins, Airflow, Visual Studio Code, IntelliJ Idea

Community Projects

Personal Blog 2023 - Present

tieukhoimai.me/insights github.com/tieukhoimai/mia-blog-v3

Beyond my professional work, I am dedicated to building and sharing knowledge with communities through my website, where I showcase my projects, ideas, and resources. In total, my website has over 10k+ views and 3.5k+ active users, with a series about Database - Data Engineering attracting 1.5k+ users and Study with Me notes having over 500 users.

Next.js | TypeScript | Tailwind CSS | MDX

MAPME - A DEDICATED INITIATIVES, AND ORGANIZATIONS TO SHAPE SUSTAINABLE SOLUTIONS 2024

mapme.asia

I've been contributing to Mapme as a data scientist, where I develop knowledge-driven maps that integrate satellite data, infrastructure maps, and local insights to promote sustainable development and enhance climate resilience in the Mekong Delta.

Google Earth Engine | Python | Geospatial Analysis

DATAcracy Program - empowering next generation of data scientists in Vietnam 2022

DTacracy

Through this program, I presented a data visualization workshop while simultaneously raising awareness among the public in Vietnam by focusing on data-driven innovations and their applications.

Tableau | Data Visualization

Awards

2023	Scholarship for Women in STEM — University of Verona
2022	Mathematical Contest in Modeling — The Consortium for Mathematics and its Applications (COMAP)
2021	Top National Students Award — Ministry of Education and Training of Vietnam
2021	Second Runner-up — Greenovator Hackathon (Bosch Vietnam)
2020	Scholarship National Program for the Development of Mathematics — VIASM
2019	Third Prize — Business Intelligence Competition
2019	Second Prize for Scientific Research Award — Ho Chi Minh City Open University
2018	Third Prize for Scientific Research Award — Ho Chi Minh City Open University
2016	Merit-based Scholarships — Ho Chi Minh City Open University